

NEXT.JS & REACT

COMPLETE COURSE SYLLABUS

Regular Batch • College / University • Corporate Training

45 Days Program • 80+ Hours Training • Hands-on Project Development

A practical React.js and Next.js development program focused on modern web application development, hands-on coding, real-world style projects, modern application architecture, APIs, Git/GitHub and interview preparation.

Course Positioning

- Regular Batch: A structured live-learning program for individuals who want to learn modern React.js and Next.js development from fundamentals through project development.
- College / University Training: Suitable for students and academic programs requiring practical web application development, guided exercises and project-based learning.
- Corporate Training: The same core curriculum can be customized according to participant experience, project requirements, technology focus and available training duration.

Course Highlights

- 45 Days Program
- 80+ Hours Training
- Hands-on React.js and Next.js development
- Real-world style application projects
- Modern web application architecture
- REST API integration
- Git & GitHub workflow
- Technical interview preparation
- Project guidance

Technology Stack

- React.js
- Next.js
- JavaScript
- HTML5
- CSS3
- REST APIs
- Git & GitHub
- Modern Web Application Development

Module 1: Web Development & JavaScript Foundations

- Modern Web Development Overview
- HTML5 fundamentals and semantic structure
- CSS3 fundamentals and responsive design
- JavaScript fundamentals
- Variables, data types and operators
- Functions and modern JavaScript syntax
- Arrays and objects
- Array methods and object manipulation
- ES6+ concepts
- Modules and reusable JavaScript code
- DOM and browser fundamentals
- Asynchronous JavaScript concepts
- Promises and async/await
- Working with JSON and APIs

Module 2: React.js Fundamentals

- Introduction to React.js
- React application structure
- Components and component-based architecture
- JSX
- Props
- State
- Event handling
- Conditional rendering
- Lists and keys
- Forms and controlled components
- Component composition
- Reusable components
- React project organization
- Hands-on React application development

Module 3: React Hooks & Application Development

- Introduction to React Hooks
- useState
- useEffect
- Effect dependencies
- Event-driven application development
- Reusable logic
- Component lifecycle concepts
- State management patterns
- Form handling and validation
- Fetching data from APIs
- Loading and error states
- Building reusable UI components
- Practical React application exercises

Module 4: React Routing & Modern Application Patterns

- Application routing concepts
- Page navigation
- Route structure
- Dynamic routes
- Nested application structures
- Reusable layouts
- Navigation patterns
- Client-side application behavior
- Application state and data flow
- Organizing larger React applications
- Practical multi-page React application

Module 5: Next.js Fundamentals

- Introduction to Next.js
- Why Next.js for modern web applications
- Next.js project structure
- App Router concepts
- Pages and layouts
- Nested layouts
- Routing
- Dynamic routes
- Static and dynamic rendering concepts
- Server-side concepts
- Client Components
- Server Components
- Loading and error handling
- Metadata and application configuration

Module 6: Next.js Application Development

- Building production-style Next.js applications
- Reusable layouts and components
- Dynamic pages
- Data fetching
- Server-side data handling
- Client-side interaction
- Forms and user input
- API integration
- REST API consumption
- JSON data handling
- Error handling
- Loading states
- Application navigation
- Responsive application development

Module 7: API Integration & Full-Stack Concepts

- REST API fundamentals
- HTTP methods
- GET, POST, PUT and DELETE
- Request and response handling
- JSON
- Connecting React/Next.js applications to APIs
- Displaying API data
- Submitting application data
- Handling API errors
- Loading and validation states
- Basic backend/API concepts
- Frontend and backend communication
- Practical API-based application

Module 8: UI, Responsive Design & Application Architecture

- Modern UI development
- Responsive web design
- Reusable UI components
- Page layout design
- Navigation and application structure
- Component organization
- Maintainable project structure
- Application architecture concepts
- User experience considerations
- Production-style frontend practices
- Building responsive real-world interfaces

Module 9: Git, GitHub & Deployment Workflow

- Introduction to Git
- Repository creation
- Git basic commands
- Commit and version history
- Branches and basic workflow
- GitHub repositories
- Pushing projects to GitHub
- Collaborative development concepts
- Environment configuration concepts
- Build and deployment concepts
- Deploying modern web applications
- Basic production troubleshooting

Module 10: Real-World Projects & Interview Preparation

- Project planning and requirement understanding
- Application architecture planning
- Component planning
- Page and route planning
- API integration in projects
- Responsive UI development
- Debugging and troubleshooting
- Code organization and refinement
- Project presentation
- Technical interview questions
- Scenario-based React.js questions
- Scenario-based Next.js questions
- JavaScript interview preparation
- Project-based interview discussion
- Resume and project explanation guidance

Practical Project Work

The course emphasizes guided application development. Learners progressively build web application features and apply React.js and Next.js concepts in practical projects.

- Responsive multi-page web application
- Reusable React component development
- Next.js application with layouts and routing
- Dynamic pages and API integration
- Forms and user interaction
- REST API-based data display
- GitHub-based project versioning
- Production-style project structure
- Final project development and presentation

Learning Outcomes

- Develop modern web applications using React.js.
- Build production-style applications using Next.js.
- Create reusable components and structured application layouts.
- Work with routing, dynamic pages, forms and application state.
- Integrate REST APIs and work with JSON data.
- Understand modern React and Next.js application architecture.
- Use Git and GitHub for project version control.
- Develop and present practical projects.
- Prepare for technical interviews using practical React.js, Next.js and JavaScript scenarios.

Delivery Options

Audience	Positioning	Suggested Delivery
Regular Learners	Next.js & React Regular Batch	45 Days / 80+ Hours
Colleges / Universities	React & Next.js Development Training	Can be aligned to academic schedule
Corporate Teams	Customized React & Next.js Training	Duration and project focus can be customized

Course Completion

A Course Completion Certificate can be provided after successful completion of the training program.

Course Duration

Regular Program: 45 Days / 80+ Hours. College, university and corporate delivery can be customized according to the requirements of the institution or organization.